

## Lecture: 14 Expansion of Functions

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### Mean Value Theorem

If  $f(x)$  is continuous in the closed interval  $[a, b]$ , and differentiable in  $(a, b)$  then there must be a value (say  $x_1$ ) of  $x$ , between  $a$  and  $b$ , such that

$$f(b) - f(a) = (b - a)f'(x_1).$$

If we put  $b - a = h$ , then  $a + \theta h$  is equal to  $a$ , if  $\theta = 0$  and is equal to  $b$  if  $\theta = 1$ . Hence,  $a + \theta h$ , where  $0 < \theta < 1$ , means some value between  $a$  and  $b$ . So the above theorem can be written as

$$f(a + h) - f(a) = hf'(a + \theta h) \text{ or } f(a + h) = f(a) + hf'(a + \theta h).$$

### Taylor's Theorem (Finite Form)

Let  $f : [a, a + h] \rightarrow \mathbb{R}$  be a function such that

(i)  $f^{n-1}$  is continuous on  $[a, a + h]$ , and

(ii)  $f^{n-1}$  is differentiable on  $(a, a + h)$ .

Then, there exists a real number  $\theta$  satisfying  $0 < \theta < 1$ , such that

$$f(a + h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{n-1}(a) + \frac{h^n(1-\theta)^{n-p}}{p(n-1)!}f^n(a + \theta h), \quad (1)$$

where  $p$  is a positive integer  $\leq n$ .

*Proof:* Since,  $f^{n-1}$  is continuous on  $[a, a + h]$ ,  $f, f', f'', \dots, f^{n-1}$  are all continuous on  $[a, a + h]$ .

Since,  $f^{n-1}$  is differentiable on  $(a, a + h)$ ,  $f, f', f'', \dots, f^{n-1}$  are all differentiable on  $(a, a + h)$ .

Let us consider the function  $\phi : [a, a + h] \rightarrow \mathbb{R}$ , defined by

$$\phi(x) = f(x) + (a + h - x)f'(x) + \frac{(a + h - x)^2}{2!}f''(x) + \dots + \frac{(a + h - x)^{n-1}}{(n-1)!}f^{n-1}(x) + A(a + h - x)^p,$$

for  $x \in [a, a + h]$  where  $A$  is a constant to be determined under the condition  $\phi(a + h) = \phi(a)$ .

$\phi(a + h) = \phi(a)$  gives

$$f(a + h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{n-1}(a) + Ah^p.$$

$$\Rightarrow A = \frac{1}{h^p} \left[ f(a+h) - f(a) - hf'(a) - \dots - \frac{h^{n-1}}{(n-1)!} f^{n-1}(a) \right]$$

$\phi$  is continuous on  $[a, a+h]$ , since  $f, f', f'', \dots, f^{n-1}$  are continuous on  $[a, a+h]$ ,  $\phi$  is differentiable on  $(a, a+h)$ , since  $f, f', f'', \dots, f^{n-1}$  are differentiable on  $(a, a+h)$ . Also  $\phi(a) = \phi(a+h)$ . By Rolle's theorem, there exists a real number  $\theta$  satisfying  $0 < \theta < 1$  such that  $\phi'(a+\theta h) = 0$ .

$$\begin{aligned} \phi'(x) &= f'(x) + [-f'(x) + (a+h-x)f''(x)] + \left[ -(a+h-x)f''(x) + \frac{(a+h-x)^2}{2!} f'''(x) \right] \\ &+ \dots + \left[ -\frac{(a+h-x)^{n-2}}{(n-2)!} f^{n-1}(x) + \frac{(a+h-x)^{n-1}}{(n-1)!} f^n(x) \right] - pA(a+h-x)^{p-1} \\ &= \frac{(a+h-x)^{n-1}}{(n-1)!} f^n(x) - pA(a+h-x)^{p-1}. \end{aligned}$$

$$\phi'(a+\theta h) = 0 \text{ gives } \frac{h^{n-1}(1-\theta)^{n-1}}{(n-1)!} f^n(a+\theta h) = pAh^{p-1}(1-\theta)^{p-1}$$

$$\text{or, } A = \frac{h^{n-p}(1-\theta)^{n-p}}{p(n-1)!} f^n(a+\theta h).$$

Therefore, we have

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \dots + \frac{h^{n-1}}{(n-1)!} f^{n-1}(a) + \frac{h^n(1-\theta)^{n-p}}{p(n-1)!} f^n(a+\theta h), 0 < \theta < 1.$$

The last term  $\frac{h^n(1-\theta)^{n-p}}{p(n-1)!} f^n(a+\theta h)$  is called the remainder after  $n$ -terms and it is denoted by  $R_n$ .  $R_n$  in this form is called the Schlomilch-Roche's form of remainder.

If  $p = 1$ ,  $R_n = \frac{h^n(1-\theta)^{n-1}}{(n-1)!} f^n(a+\theta h)$ .  $R_n$  in this form is called the Cauchy's form of remainder.

If  $p = n$ ,  $R_n = \frac{h^n}{n!} f^n(a+\theta h)$ .  $R_n$  in this form is called the Lagrange's form of remainder.

Taylor's theorem is more usually written in the following form

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots + \frac{(x-a)^{n-1}}{(n-1)!} f^{n-1}(a) + \frac{(x-a)^n}{n!} f^n\{a+\theta(x-a)\}.$$

Putting  $a = 0$ , we get Maclaurin's expansion as:

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^{n-1}}{(n-1)!} f^{n-1}(0) + \frac{x^n}{n!} f^n(\theta x).$$

**Example.** Let  $a \in \mathbb{R}$  and a real function  $f$  defined on some neighborhood  $N(a)$  of  $a$  be such that  $f''$  is continuous at  $a$  and  $f''(a) \neq 0$ . Prove that  $\lim_{h \rightarrow 0} \theta = \frac{1}{2}$ , where  $\theta$  is given by  $f(a+h) = f(a) + hf'(a+\theta h)$  ( $0 < \theta < 1$ ).

**Proof.** Since,  $f''$  is continuous at  $a$ ,  $f''$  exists in some neighborhood  $(a-\delta, a+\delta) \subset N(a)$ .

**Case 1.** Let  $0 < h < \delta$ .

By Taylor's theorem with Lagrange's form of remainder after 2 terms, we have

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a+\theta'h), 0 < \theta' < 1.$$

Therefore,  $f'(a+\theta h) = f'(a) + \frac{h}{2} f''(a+\theta'h)$ .

Applying Mean value theorem to  $f'$  on the interval  $[a, a+\theta h]$ , we have

$$f'(a+\theta h) = f'(a) + \theta h f''(a+\theta\theta''h), 0 < \theta'' < 1.$$

$$\text{Therefore, } \frac{1}{2} f''(a+\theta'h) = \theta f''(a+\theta\theta''h).$$

Taking limit as  $h \rightarrow 0$ , we have

$$\frac{1}{2} \lim_{h \rightarrow 0+} f''(a+\theta'h) = \lim_{h \rightarrow 0+} [\theta f''(a+\theta\theta''h)].$$

Since,  $f''$  is continuous at  $a$ ,

$$\lim_{h \rightarrow 0+} f''(a+\theta'h) = f''(a), \lim_{h \rightarrow 0+} f''(a+\theta\theta''h) = f''(a).$$

Therefore,

$$\lim_{h \rightarrow 0+} \theta = \frac{1}{2}, \text{ since } f''(a) \neq 0.$$

**Case 2.** Let  $-\delta < h < 0$ . Proceeding similarly, we have  $\lim_{h \rightarrow 0-} \theta = \frac{1}{2}$ .

$$\text{Hence, } \lim_{h \rightarrow 0} \theta = \frac{1}{2}.$$

### Taylor's Infinite Series

Let  $a \in \mathbb{R}$ . Let a real function  $f$  defined on some neighborhood  $N(a)$  of  $a$  be such that  $f^{n-1}$  is differentiable on  $N(a)$ . Then, for any  $x \in [N(a)-a]$ ,  $f(x) = P_n(x) + R_n(x)$ , where  $R_n(x)$  is the remainder after  $n$  terms and

$$P_n(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots + \frac{(x-a)^{n-1}}{(n-1)!} f^{n-1}(a).$$

$P_n(x)$  is a polynomial of degree  $n - 1$ , such that

$$P_n(a) = f(a), P'_n(a) = f'(a), P''_n(a) = f''(a), \dots, P_n^{n-1}(a) = f^{n-1}(a).$$

$P_n(x)$  is called the  $n$ th Taylor polynomial of  $f$  about the point  $a$ .

If  $f$  be a function, such that for all  $n \in \mathbb{N}$ ,  $f^n$  exists on  $N(a)$ , then the polynomial  $P_n(x)$  for any  $x \in N'(a)$  takes the form of an infinite series

$$f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2!}f''(a) + \dots$$

Since,  $f(x) = P_n(x) + R_n(x)$  for any  $n$ , the above series will converges to  $f(x)$  if and only if

$$\lim_{n \rightarrow \infty} R_n(x) = 0.$$

Therefore, if

$$\lim_{n \rightarrow \infty} R_n(x) = 0,$$

$$f(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2!}f''(a) + \dots$$

In particular, if  $a = 0$ , then Taylor's infinite series for the function  $f$  about 0 takes the form  $f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots$ , this is called Maclaurin's infinite series for  $f$ .

**Example.** Expand  $e^x$  by Taylor's theorem.

**Solution.** For all  $n \in \mathbb{N}$ ,  $f^n(x) = e^x, x \in \mathbb{R}$ .

By Taylor's theorem with Lagrange's for of remainder after  $n$  terms, for a non zero  $x \in \mathbb{R}$ ,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \dots + \frac{x^{n-1}}{(n-1)!}f^{n-1}(0) + R_n(x),$$

where  $R_n(x) = \frac{x^n}{n!}f^n(\theta x)$  for some real number  $\theta$  satisfying  $0 < \theta < 1$ .

Since, for all  $n \in \mathbb{N}$ ,  $f^n(x)$  exists for all real  $x$ , the right hand polynomial takes the form of an infinite series as  $n \rightarrow \infty$ . The infinite series will converge to  $f(x)$  for those non zero real  $x$  for which  $\lim_{n \rightarrow \infty} R_n = 0$ .

$$R_n = \frac{x^n}{n!}e^{\theta x}.$$

Let  $u_n = \frac{|x|^n}{n!}, x \neq 0$ . Then

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0.$$

This proves  $\lim u_n = 0$ .

For all real  $x$ ,  $e^{\theta x}$  is bounded. Hence,  $\lim_{n \rightarrow \infty} |R_n| = 0$  for all real  $x \neq 0$  and this implies  $\lim R_n = 0$  for all real  $x \neq 0$ .

Consequently, the infinite series  $1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots$  converges to  $e^x$  for all real  $x \neq 0$ .

At  $x = 0$ , the convergence holds trivially.

So,  $e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots$  for all  $x \in \mathbb{R}$ .

Reference Book: Differential Calculus, Dr. Gorakh Prasad. Pothishala Private LTD.